

Fields of the Planet: Field Boundary Mapping Beyond 10m

Supplementary Material

A. Polygon-metric definitions

All object-level metrics are computed on vector geometries for the field-interior class. Predicted masks are vectorized into polygons and scored against the original FTW ground-truth polygons; pixel IoU is computed before vectorization and reported only for continuity with the released PRUE protocol.

Vectorization and matching. For each patch, the post-processed field-interior prediction is converted into polygons $\mathcal{P} = \{p_1, \dots, p_M\}$ and compared with the ground-truth polygons $\mathcal{G} = \{g_1, \dots, g_N\}$. All intersections and unions use polygon areas from the vector geometries. At IoU threshold τ , a predicted polygon and a ground-truth polygon are matched when

$$\text{IoU}(g, p) = \frac{\text{area}(g \cap p)}{\text{area}(g \cup p)} > \tau. \quad (1)$$

For $\tau \geq 0.5$, disjoint polygon interiors make the match set $\mathcal{M}_\tau$ a partial one-to-one matching. We then define $\text{TP}_\tau = |\mathcal{M}_\tau|$, $\text{FP}_\tau = M - \text{TP}_\tau$, and $\text{FN}_\tau = N - \text{TP}_\tau$.

Object F1 and panoptic quality. Object F1 is reported at $\text{IoU} = 0.5$ and over the COCO threshold grid $\mathcal{T} = 0.5, 0.55, \dots, 0.95$ [5]:

$$\text{F1}_{[.5:.95]} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \text{F1}_\tau. \quad (2)$$

Panoptic quality follows Kirillov et al. [4]. With matches fixed at $\tau = 0.5$, recognition quality is $\text{RQ} = \text{F1}_{0.5}$, segmentation quality is the mean IoU of matched polygons,

$$\text{SQ} = \frac{1}{\text{TP}_{0.5}} \sum_{(g,p) \in \mathcal{M}_{0.5}} \text{IoU}(g, p), \quad (3)$$

and $\text{PQ} = \text{SQ} \times \text{RQ}$. If $\text{TP}_{0.5} = 0$, we set $\text{SQ} = 0$ and $\text{PQ} = 0$; matched-boundary error is undefined.

Polygon-count error. Let $\bar{M}$ and $\bar{N}$ denote the mean numbers of predicted and ground-truth polygons per patch within a country. We report the normalized count discrepancy

$$|\Delta N|/N = \frac{|\bar{M} - \bar{N}|}{\bar{N}}, \quad (4)$$

computed per country and then macro-averaged. This is a coarse object-count check independent of IoU matching; it measures count mismatch magnitude but not whether errors come from over- or under-prediction.

Boundary error. For each matched pair $(g, p) \in \mathcal{M}_{0.5}$, we compute a symmetric boundary chamfer distance between the one-pixel-wide ground-truth and predicted boundary sets on the evaluated raster grid. Nearest-boundary distances are read from Euclidean distance transforms, averaged in both directions, and converted to meters by multiplying by the evaluated grid's ground sample distance: 3m for PlanetScope and 10m for native Sentinel-2. We report the mean and 95th percentile over matched pairs. Boundary error therefore measures localization for fields that were recovered; missed and hallucinated fields are penalized through RQ and object F1.

Pixel IoU. For continuity with PRUE [7], we also report the pixel-level Jaccard index for the field-interior class,

$$\text{IoU}_{\text{px}} = \frac{|\hat{Y}_{\text{field}} \cap Y_{\text{field}}|}{|\hat{Y}_{\text{field}} \cup Y_{\text{field}}|}, \quad (5)$$

computed on raster masks before vectorization. Unlike the polygon metrics, pixel IoU has no notion of individual parcels.

Aggregation. Metrics are first pooled within each country: TP/FP/FN counts and matched IoUs are accumulated across patches before computing RQ, SQ, PQ, and $F1_{[.5:.95]}$, while chamfer is averaged over that country’s matched pairs. Main-paper Table 1 reports macro-averages over the ten dense-label held-out countries. If a configuration produces no matched polygons in a country, boundary error is undefined for that country and excluded from the boundary-error average; all other columns remain defined.

Field-area breakdown. For main-paper Table 3, ground-truth polygons are assigned by UTM area A to *small* ($A < 0.5$ ha), *medium* ($0.5 \leq A < 2$ ha), or *large* ($A \geq 2$ ha). The 2 ha threshold follows the standard smallholder ceiling [6], while 0.5 ha separates the dominant sub-hectare fields in FTW. Matched true positives and false negatives are binned by ground-truth area; unmatched predictions are counted as false positives in the bin determined by their predicted area. PQ is reported per bin and macro-averaged over the ten dense-label held-out countries.

B. Upsampled Sentinel-2 Control

This control separates output-grid resolution from image resolution. We give the Sentinel-2 model a finer 512-pixel prediction grid without adding new image information, either by upsampling at evaluation time or by training on bilinearly upsampled Sentinel-2 imagery. Both settings improve object-level polygon scores, but neither recovers the meter-scale boundary precision of native 3m PlanetScope imagery.

Table 4. **Upsampling Sentinel-2 improves object recovery but not boundary precision.** Results use B3 models on the dense held-out split with WS + TTA. Rows share the same upsampled-comparable scoring protocol.

Sentinel-2 grid	PQ	SQ	RQ	Bd. err (m)
Native (256, 10m)	23.8	67.8	30.8	18.6
Upsampled at eval (512)	27.4	75.8	34.9	31.3
Trained upsampled (512)	31.5	77.5	39.5	28.0
Planet (512, native 3m)	36.0	77.9	45.2	7.4

C. Dataset scope and per-tile UDM2 quality

Per-region usable-patch rates (clear $\geq 95\%$, unusable $\leq 5\%$) vary widely across the release; smallholder regions (e.g. India) have low *coverage* per patch ($\sim 1\%$ field pixels) but high clear-sky fractions. The full per-region index ships with the release as a CSV.

Table 5. **Dataset scope.** FTP pairs FTW’s 24 countries / 25 labeled regions with two seasonal PlanetScope windows per patch. *Success rate* denotes the fraction of FTW patch-window targets with a successfully extracted PlanetScope image.

Property	Value
Geographic coverage	
Countries / regions / continents	24 / 25 / 4
FTW patches	70,484
(Patch, window) targets	140,968
Source imagery	
Unique PSScene COGs touched	6,113
Mean windows per scene	21.8
Released artifacts	
PlanetScope SR (4-band, uint16)	133,168
Per-window UDM2 QA statistics	129,490
3-class labels (uint8, 1/patch)	66,584
Yield	
Successfully matched pairs	133,168
Success rate	94.5%
Total dataset size	102 GB

Table 6. **Per-tile UDM2 coverage.** Percentage of patch-window tiles whose UDM2 class coverage exceeds each threshold. Most tiles are clear (95.9% are $\geq 50\%$ clear); tiles with $\geq 50\%$ cloud, shadow, haze, or unusable pixels are resampling candidates.

Band	any (>0)	$\geq 50\%$	$\geq 90\%$
clear	98.9	95.9	91.1
cloud	9.0	1.0	0.2
shadow	6.9	0.2	0.0
light_haze	5.1	2.0	1.1
unusable	11.3	1.3	0.4

% of (patch, window) tiles; each tile’s class area as a fraction of the window.

D. Held-out post-processing sweep

Table 7 reports the full watershed/D4-TTA sweep for PRUE+ models on the ten-country dense held-out set. The released FTW-PRUE rows are included as external reference values from the original PRUE test split, not as held-out macro-averages.

PRUE+ augmentation settings. PRUE+ is implemented with Kornia [8]. It keeps the released PRUE preprocessing augmentations—`preprocess_aug` with divisor sampled from [5000, 15000] and `RandomResizedCrop` with scale 0.3–0.9, ratio 0.75–1.33, and $p = 0.5$ —then adds seasonal-window swap and per-band gamma jitter ($\gamma \sim \mathcal{U}[0.8, 1.2]$, $p = 0.3$). The final geometry/noise bundle adds affine intensity jitter ($a \sim \mathcal{U}[0.9, 1.1]$, $b \sim \mathcal{U}[-0.02, 0.02]$, $p = 0.3$), Gaussian blur (3×3 , $\sigma \sim \mathcal{U}[0.5, 1.5]$, $p = 0.3$), Gaussian noise ($\sigma = 0.015$, $p = 0.3$), single-window dropout ($p = 0.15$), rotation ($\pm 30^\circ$, $p = 0.5$), shear ($\pm 5^\circ$, $p = 0.3$), and boundary-thickness jitter by random binary dilation of 0–2 pixels ($p = 0.5$).

Table 7. **Held-out post-processing sweep.** Object F1 for B3/B7 PRUE+ models across watershed (WS) and D4 test-time augmentation (TTA) settings. Dense-label macro-averages exclude presence-only Kenya and retain Portugal; released FTW-PRUE rows are published PRUE reference values. Best per row is bolded.

Model	Backbone	Obj F1 (10-country dense held-out)				Pix IoU
		no-WS	no-WS	WS	WS	
		no-TTA	TTA	no-TTA	+TTA	
<i>FTW-PRUE, released by [7]</i>						
FTW-PRUE	B3	—	—	—	43.0 [‡]	74.0 [‡]
FTW-PRUE	B5	—	—	—	46.0 [‡]	75.0 [‡]
FTW-PRUE	B7	—	—	—	47.0 [‡]	76.0 [‡]
<i>Ours — FTP-PRUE+</i>						
FTP-PRUE+	B3	43.5	43.9	44.1	45.2	68.8
<i>FTW-PRUE baselines re-trained with our PRUE+ recipe</i>						
FTW-PRUE+	B3	33.4	33.3	33.9	34.9	61.8
FTW-PRUE+	B7	39.9	39.5	40.4	40.7	63.6

Table 8. **Recipe ablation summary.** Object-F1 changes on the ten-country dense held-out set, using WS + TTA unless noted. Checkmarks indicate components kept in FTP-PRUE+; × marks rejected variants.

Lever	Use	Δ ObjF1
Augmentation & training		
PRUE preprocess+resize aug. [7]	✓	+2.8
swap_order + per-band γ	✓	+1.0
Geometry/noise bundle (PRUE+: affine, blur, noise, window dropout, rot, shear, jitter)	✓	+4.5
Backbone B3 → B7 (no gain, larger)	×	~0
Soft cIDice [9] on boundary	×	failed
SDF auxiliary head [1, 3] (with PRUE+)	×	-3.1
Frame-field head [2]	×	-1.7
CutMix [10] (2 px ignore)	×	-0.1
Inference & post-processing		
Marker-controlled watershed	✓	+0.5
D4 8-way TTA (with PRUE+)	✓	+0.9
Checkpoint selection		
Last vs best-by-val checkpoint	either	≤ 0.3

References

- [1] Benjamin Bischke, Patrick Helber, Joachim Folz, Damian Borth, and Andreas Dengel. Multi-task learning for segmentation of building footprints with deep neural networks. In *IEEE International Conference on Image Processing (ICIP)*, 2019. 3
- [2] Nicolas Girard, Dmitriy Smirnov, Justin Solomon, and Yuliya Tarabalka. Polygonal building segmentation by frame field learning. In *CVPR*, 2021. 3
- [3] Zeeshan Hayder, Xuming He, and Mathieu Salzmann. Boundary-aware instance segmentation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 3
- [4] Alexander Kirillov, Kaiming He, Ross Girshick, Carsten Rother, and Piotr Dollár. Panoptic segmentation. In *CVPR*, 2019. 1
- [5] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014. 1

- 073 [6] Sarah K. Lowder, Jakob Skoet, and Terri Raney. The number, size, and distribution of farms, smallholder farms, and family farms
074 worldwide. *World Development*, 87:16–29, 2016. 2
- 075 [7] Gedeon Muhawenayo, Caleb Robinson, Subash Khanal, Zhanpei Fang, Isaac Corley, Alexander Wollam, Tianyi Gao, Leonard
076 Strnad, Ryan Avery, Lyndon Estes, Ana M. Tárano, Nathan Jacobs, and Hannah Kerner. PRUE: A practical recipe for field boundary
077 segmentation at scale. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages
078 6484–6495, 2026. 1, 3
- 079 [8] Edgar Riba, Dmytro Mishkin, Daniel Ponsa, Ethan Rublee, and Gary Bradski. Kornia: an open source differentiable computer vision
080 library for PyTorch. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2020. 3
- 081 [9] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich
082 Bauer, and Bjoern H. Menze. clDice - a novel topology-preserving loss function for tubular structure segmentation. In *CVPR*, 2021. 3
- 083 [10] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization strategy
084 to train strong classifiers with localizable features. In *ICCV*, 2019. 3